A Physiologically Interpretable Multimodal Model for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

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Abstract

A clinical polysomnogram records far more than the brain. Alongside the electroencephalogram it captures eye movements, muscle tone, the electrocardiogram, nasal and oral airflow, thoracic and abdominal breathing effort, and blood oxygen saturation. In patients recovering from ischemic stroke, a group in which breathing is disturbed during sleep at high rates, these cardiorespiratory signals carry the disease itself. We build one model that reads the full polysomnogram and produces two clinical outputs at once: the sleep stage of every 30-second epoch and a per-epoch label for respiratory events. The model encodes the neural and the cardiorespiratory signals in two parallel streams, fuses them, follows the night with a bidirectional recurrent decoder, and drives a staging head and a respiratory head, with a direct connection that carries the cardiorespiratory signal to the respiratory head so oxygen desaturation reaches the decision at full strength. We benchmark the model against deep-learning and classical baselines on iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke (99 patients), all under patient-independent ten-fold cross-validation. The single model stages sleep at 0.722 accuracy with Cohen's κ of 0.611, at the accuracy ceiling this cohort imposes on every model family, and it detects respiratory events at

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0.711 area under the ROC curve, a capability none of the staging models possess. The respiratory read-out is grounded in physiology: when we remove the airflow channel from which the events were scored, detection holds at 0.704, because the model reads oxygen desaturation, breathing effort, cardiac variability, and the cortical arousals that follow an obstructed breath. Each modality contributes where its physiology lives. Code and features are released.

Keywords: Polysomnography, multimodal learning, sleep staging, sleep apnea, ischemic stroke, feature fusion, multi-task learning, cardiorespiratory signals.

CONTRIBUTIONS. **Md Wahiduzzaman Suva** (Conceptualization, Methodology, Software, Investigation, Validation, Formal analysis, Writing – original draft, Project administration): conceived the joint formulation and led the study; designed and implemented MM-Net, the raw multimodal CNN and the ten-fold reproduction engine; ran the joint staging–respiratory experiments, the modality-ablation grid and the significance tests; led the verification pass; produced Tables III–VI and wrote Sections I, V–VIII. **Esm-e Moula Chowdhury Abha** (Investigation, Software, Validation, Visualization, Writing – original draft): trained and evaluated the deep comparison models – DeepSleepNet, AttnSleep and the healthy-pretrained transfer model – on matched folds; produced the architecture diagram (Fig. 3) via the draw.io vector pipeline; wrote Sections II–III; produced Table VIII and compiled the bibliography. **Md Imtiaj Alam Sajin** (Data curation, Software, Investigation, Visualization): built the polysomnography preprocessing pipeline – channel mapping, resampling, epoching – and the feature extractors; trained the respiratory reference detectors; produced Figures 4–10 and Tables I, II and VII; wrote Section IV. **Md Iftekharul Mobin** (Supervision, Project administration, Conceptualization, Writing – review & editing, Resources): supervised the project and its clinical framing, critically revised the manuscript, and provided institutional resources; corresponding author.

1. Introduction

Sleep is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have moved from convolutional and recurrent encoders [2, 3, 4] through attention and sequence models [5, 6, 7] to large self-supervised EEG foundation models [8, 9, 10, 11]. On healthy sleepers these methods reach roughly 85% accuracy, and recent reviews argue the field is near a ceiling set by inter-scorer disagreement [12].

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral oxygen saturation (SpO_2). These cardiorespiratory channels are where sleep-disordered breathing writes itself: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient's sleep. We use the full polysomnogram, and we ask a question that the staging race does not: what does a model gain, and on which clinical task, when it reads the whole recording rather than the brain alone.

We study this on iSLEEPS [13], released in 2026 as the first public PSG corpus dedicated to subacute ischemic stroke. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [14, 15], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 77 of 96 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most. The published staging baselines on this corpus reach 74.7% accuracy for an LSTM, about ten points below the healthy state of the art, and a concurrent study reports that deep networks attend to non-physiological regions on this cohort [16], so the setting also stresses every staging architecture we can bring to it.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem, and we place it inside a broad benchmark on the same folds. Our contributions are:

- **C1. One model for two clinical read-outs from the full polysomnogram.** It stages sleep at 0.722 accuracy (κ 0.611) and, from the same forward pass, detects respiratory events at 0.711 AUC, with a direct cardiorespiratory path to the respiratory head so oxygen-desaturation cues reach the decision undiminished (Section 5).

- **C2. Healthy-sleep deep models fail on the injured brain.** Deep networks trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal collapse to 0.61–0.69 accuracy on this cohort (Table 3), while classical pipelines saturate below 0.75; the compact feature-based model reaches 0.722 and, uniquely, also produces a respiratory read-out. Staging accuracy here reflects a cohort ceiling; the joint respiratory output is what we contribute.

C3. A physiologically grounded respiratory read-out. Removing the airflow channel from which the events were scored leaves detection at 0.704 AUC, because the decision rests on oxygen desaturation, breathing effort, cardiac variability, and cortical arousals (Section 6.7).

C4. Causal, per-modality attribution. A leave-one-out modality-ablation grid shows each channel carries the task its physiology governs: removing the cardiorespiratory stream lowers only respiratory detection (Wilcoxon $p = 0.004$), while removing the ocular channel lowers only staging, giving a falsifiable and physiologically interpretable account of the model (Table 5).

Together, these results reframe the problem on this cohort: the useful gains come from reading the whole sleeping body and attributing each output to the physiology that produces it; deeper single-channel staging does not help on this cohort. The remainder of the paper develops the model (Section 5) and tests each claim in turn.

2. Related Work

2.1. Single-channel deep sleep staging

Automated staging matured on one EEG channel. DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep introduced attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4, 7]; and XSleepNet fused raw and spectral views [6]. More recent backbones push this further: SleepTransformer adds interpretability and uncertainty to an attention model [17], L-SeqSleepNet models whole-night cycles [18], ZleepAnlystNet and FlexSleepTransformer explore separated training and flexible channel sets [19, 20], and linear-time state-space backbones reach competitive accuracy [21, 22]. These models set the accuracy on healthy corpora but depend on the cortical micro-events that stroke disturbs, which is why they transfer poorly to patients [16].

2.2. Multi-channel and foundation models

Later work reads several PSG channels. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [23], and self-supervised EEG foundation models such as LaBraM, CBraMod, EEGPT, and BIOT learn transferable representations from large unlabeled corpora [8, 9, 10, 11]. SleepFM extends this idea to several PSG modalities [24], while cross-scale foundation models and standardized benchmarks continue to appear [25, 26, 27], self-supervised pretraining is now systematically evaluated for label efficiency [28], transfer and domain adaptation improve robustness across cohorts [29], and recent surveys and clinical benchmarks catalogue the field [30, 31]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Reported gains from adding respiratory channels to *staging* are small and cohort-dependent; our benchmark quantifies this directly on a high-burden clinical cohort and separates the staging effect from the respiratory one.

2.3. Detection of sleep-disordered breathing

A parallel line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is a defining marker of an event [32]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We instead detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

2.4. Sleep and breathing after stroke

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [14, 15, 33]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [34, 35, 36, 37, 38]; quantitative EEG markers such as the brain symmetry index further track hemispheric injury [39]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population. Sleep staging on ischemic stroke itself is nascent, reported only as baselines on the iSLEEPS corpus [13], and no prior method addresses staging and respiratory detection jointly, from the full polysomnogram, on a stroke population; we make this comparison quantitative in Table 9.

3. Critical Gaps and Limitations of Prior Work

Three gaps run through the literature above, and each maps onto one of our contributions. *(i) Single-task, single-signal design.* Almost every staging model reads one or a few EEG channels and predicts only the stage, while apnea detectors read breathing signals and predict only events; the two literatures rarely meet in one model, leaving the clinician with two separate tools. Contribution C1 answers this with a single model that reads the full polysomnogram and produces both outputs. *(ii) Evaluation on healthy sleepers.* Staging accuracy is reported almost entirely on healthy corpora such as Sleep-EDF [40, 41], where deep and foundation models reach a ceiling, while performance on focal brain injury is largely unmeasured. Contribution C2 benchmarks strong deep architectures, classical gradient-boosting pipelines [42, 43, 44], and graph models [45] on a subacute-stroke cohort, and shows that the healthy-built networks collapse. *(iii) Attribution, reliability, and cost.* Interpretability is usually post-hoc and unfalsifiable, predictive uncertainty is seldom modeled [46], and the compute and energy footprint of heavy models is rarely reported [47]. Contributions C3 and C4 answer this with a physiologically grounded, non-circular respiratory read-out and a causal modality-ablation that attributes each output to the channel its physiology governs, in a compact model.

4. Dataset and Preprocessing

iSLEEPS [13] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments, giving $N = 99$. Sleep staging uses all 99 patients; the respiratory task uses the $N = 96$ with complete cardiorespiratory channels, and we state which N applies to each reported number. The corpus totals 89,532 epochs. The stage distribution (Table 1) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

Channels we read, and channels we do not. A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2),

Table 1: Stage distribution and respiratory-event prevalence in the corpus we model ($N = 99$).

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

Respiratory events: 16.0% of epochs positive; median 13% per patient (range 0 to 60.5%); 77 of 96 patients above 5%.

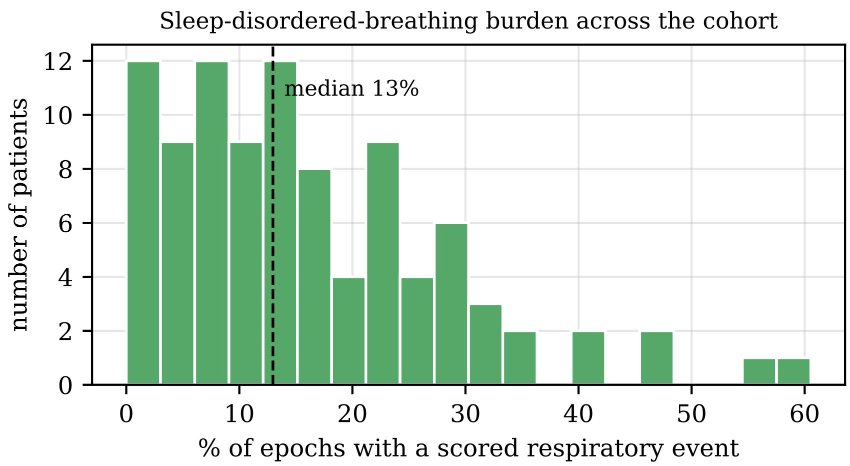

Figure 1: Sleep-disordered-breathing burden across the cohort. Each bar counts patients by the fraction of epochs that carry a scored respiratory event. The median patient sits at 13% and many exceed 30%, so the cardiorespiratory signal is abundant.

the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, SpO_2, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the SpO_2 and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device housekeeping, not physiology. All neural channels are resampled to 100 Hz and the cardiorespiratory channels to a common 25 Hz grid; SpO_2, recorded natively at 4 Hz, is upsampled to this grid, so its effective resolution remains that of the 4 Hz oximeter. Signals are read in physical units, which matters because

SpO_2 near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales. Per-epoch respiratory labels are the corpus's scored events mapped onto the 30 s grid, and an epoch is positive when it overlaps an event.

5. Proposed Method

5.1. Problem formulation

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\text{eeg}} \in \mathbb{R}^{188}$ and a cardiorespiratory feature vector $\mathbf{f}_t^{\text{car}} \in \mathbb{R}^{14}$, and we learn a subject-independent map

$$f_\theta : \left(\mathbf{f}_{1:T}^{\text{eeg}}, \mathbf{f}_{1:T}^{\text{car}}\right) \longmapsto \left(\hat{y}_{1:T}^{\text{stg}},\ \hat{y}_{1:T}^{\text{apn}}\right), \tag{1}$$

where $\hat{y}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{y}_t^{\text{apn}} \in [0,1]$ is the probability that epoch t contains a respiratory event (Fig. 2).

5.2. Physiological feature representations

Neural features (188). Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–12), sigma (12–16), and beta (16–30) Hz; the relative power in band b is

$$\mathrm{RP}_b = \frac{\int_{f\in b} P(f)\, df}{\int_{0.5}^{30} P(f)\, df}. \tag{2}$$

We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [48], and time-domain statistics. Sleep spindles are detected on the sigma-band analytic envelope $e(t) = |\mathcal{H}\{x_\sigma\}(t)|$ from the Hilbert transform $\mathcal{H}$, with density counted above a per-epoch threshold; slow-wave amplitude, ocular-movement energy on the EOG, and tonic EMG level complete the set.

Cardiorespiratory features (14). From the seven cardiorespiratory channels we compute SpO_2 mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and thoraco-abdominal asynchrony, the loss of coordination between chest and abdomen that marks an obstructed breath,

$$\mathrm{asyn}_t = \left|\, \mathrm{corr}(x_t^{\text{tho}},\ x_t^{\text{abd}})\, \right|. \tag{3}$$

Both feature sets are standardized per patient, which removes person-to-person offsets so the model generalizes across people.

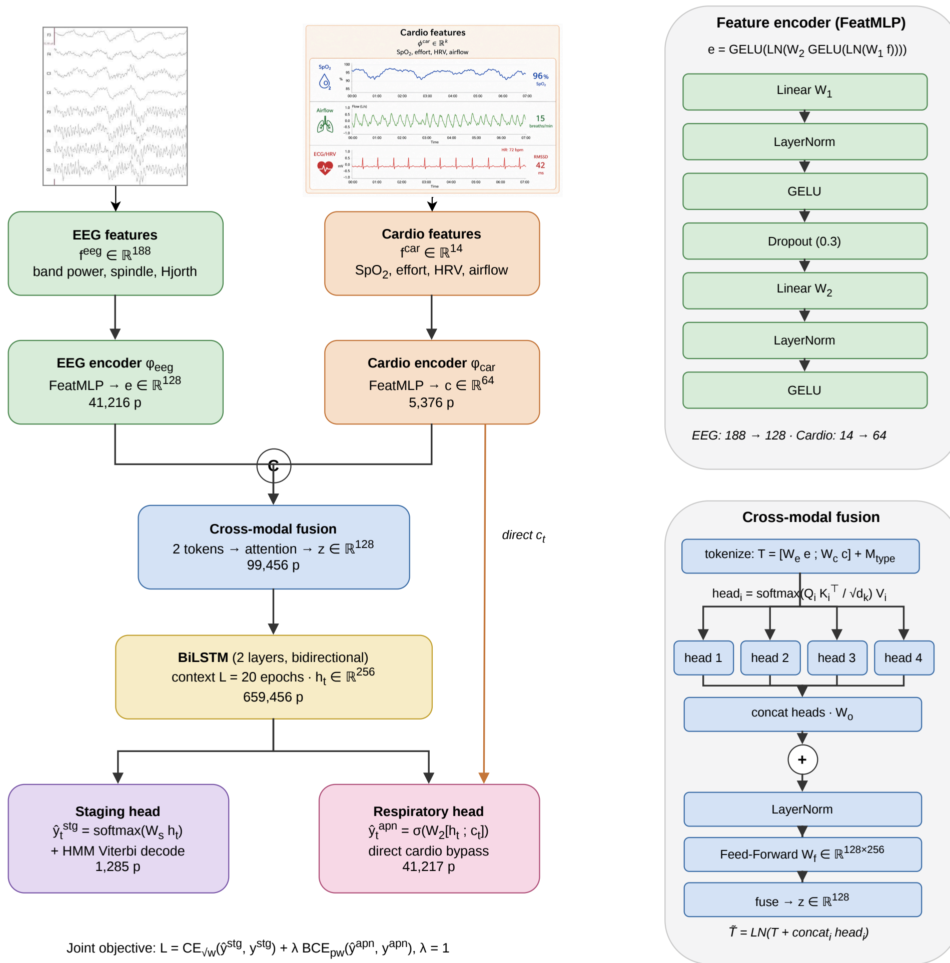

Figure 2: The two-stream, multi-task model. (A) The neural features and the cardiorespiratory features are encoded in parallel, combined by a cross-modal fusion block, and read across the night by a bidirectional LSTM. A staging head reads the recurrent state; a respiratory head reads the recurrent state together with a direct copy of the cardiorespiratory embedding, so oxygen-desaturation and effort cues reach the respiratory decision without passing through the staging-oriented fusion. (B) Detail of the feature encoder and of the fusion block, which tokenizes the two modalities, runs four-head attention, and combines the result with a residual connection.

5.3. Two-stream encoders

Each stream is a two-layer perceptron with LayerNorm,

$$e_t = \mathrm{GELU}\big(\mathrm{LN}(W_2\, \mathrm{GELU}(\mathrm{LN}(W_1 \mathbf{f}_t^{\mathrm{eeg}})))\big) \in \mathbb{R}^{128}, \tag{4}$$

$$c_t = \mathrm{GELU}\big(\mathrm{LN}(U_2\, \mathrm{GELU}(\mathrm{LN}(U_1 \mathbf{f}_t^{\mathrm{car}})))\big) \in \mathbb{R}^{64}. \tag{5}$$

LayerNorm rather than BatchNorm is used because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because 14 slow descriptors carry far less structure than 188 neural features.

5.4. Cross-modal fusion

The two embeddings become two tokens with learned modality-type embeddings $M \in \mathbb{R}^{2\times D}$, $D = 128$,

$$T = \big[\, W_e\, e_t \;;\; W_c\, c_t \,\big] + M \;\in\; \mathbb{R}^{2\times D}. \tag{6}$$

Four-head scaled dot-product attention lets each modality read the other,

$$\begin{aligned} \mathrm{head}_i &= \mathrm{softmax}\Big(\tfrac{(TW_i^Q)(TW_i^K)^\top}{\sqrt{d_k}}\Big)\, TW_i^V, \\ A &= \big[\mathrm{head}_1; \ldots; \mathrm{head}_4\big] W^O, \end{aligned} \tag{7}$$

followed by a residual connection, normalization, and a projection of the two refined tokens,

$$\tilde{T} = \mathrm{LN}(T + A), \qquad z_t = \mathrm{GELU}\big(W_f\, \mathrm{vec}(\tilde{T})\big) \in \mathbb{R}^{D}. \tag{8}$$

We also evaluate a plain concatenation in place of attention; the two perform alike (Section 6.6), so the reported gain reflects the added modality; the fusion design contributes little.

5.5. Temporal decoder and two heads

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \tag{9}$$

and the forward and backward states are concatenated into $h_t \in \mathbb{R}^{256}$. The staging head is a linear map $\hat{y}_t^{\text{stg}} = \text{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t,

$$\hat{y}_t^{\text{apn}} = \sigma\big(W_2\, \text{GELU}(W_1[h_t; c_t])\big). \tag{10}$$

This direct connection matters: without it the respiratory decision must survive a fusion trained mostly for staging, and the fall in oxygen that defines an event is weakened along the way. The full model holds 0.86 million parameters.

5.6. Training and decoding

The two tasks are trained together with equal weight, $\mathcal{L} = \mathcal{L}^{\text{stg}} + \lambda\, \mathcal{L}^{\text{apn}}$ with $\lambda = 1$, combining a class-weighted staging cross-entropy and a positive-weighted respiratory cross-entropy,

$$\mathcal{L}^{\text{stg}} = -\sum_t \sum_k w_k\, y_{t,k}^{\text{stg}} \log \hat{y}_{t,k}^{\text{stg}}, \tag{11}$$

$$\mathcal{L}^{\text{apn}} = -\sum_t \big[\pi\, a_t \log \hat{y}_t^{\text{apn}} + (1 - a_t) \log(1 - \hat{y}_t^{\text{apn}})\big]. \tag{12}$$

The staging weights use the square root of inverse class frequency, $w_k \propto \sqrt{N/(Kn_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate. At inference the staging posteriors are smoothed by a hidden-Markov Viterbi pass whose transition matrix A^{H} and prior are counted from the training labels [49],

$$\delta_t(j) = \max_i \big[\delta_{t-1}(i) + \log A_{ij}^{\text{H}}\big] + \log \hat{y}_t^{\text{stg}}(j), \tag{13}$$

recovered by back-tracking, which restores the natural persistence of sleep. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

6. Experiments

6.1. Protocol and metrics

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a

Table 2: Training configuration, shared across all folds, re-implemented baselines, and ablations.

Setting	Value
Framework / hardware	PyTorch, NVIDIA RTX 2060 (6 GB)
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
LR schedule	cosine annealing
Sequence length L	20 epochs (10 min)
Batch size	32
Max epochs / patience	45 / 8
Gradient clipping	5.0
Staging class weights	$\surd$ inverse-frequency
Respiratory pos. weight	$\pi = n_- / n_+$
Loss balance λ	1.0
Staging post-processing	HMM Viterbi smoothing
Cross-validation	10-fold, patient-independent
Random seed	42

disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K}\sum_k \mathrm{F1}_k$, and Cohen's κ. The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

6.2. Implementation and hyperparameters

The model is implemented in PyTorch and trained on a single NVIDIA RTX 2060 GPU (6 GB) under Windows, with all randomness seeded to 42 so that runs are deterministic and reproducible. Table 2 lists the training configuration, which is identical across all folds, all baselines we re-implement, and every ablation, so that reported differences come from the model and not from tuning.

6.3. Compared Methods

We compare the model against deep networks evaluated on the same folds: convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN–BiLSTM), a network transferred from

Table 3: Sleep staging on iSLEEPS (ten-fold, patient-independent): the proposed feature-based model against deep networks trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal. "feat" marks engineered features; best per column in **bold**. Deep architectures built for healthy sleep lose 15–20 points on this stroke cohort, while the compact feature-based model is the strongest here and the only one that also detects respiratory events (Table 5). Classical feature and ensemble pipelines, which reach a comparable staging ceiling, are summarised in the text.

	Model	Input	Params↓	Acc↑	Macro-F1
Deep networks	CNN-ResNet18 [13]	1 EEG	∼11 M	0.617	0.544
	DeepSleepNet [2]	1 EEG	∼21 M	0.615	0.567
	AttnSleep [5]	1 EEG	∼0.5 M	0.686	0.629
	CNN + BiLSTM	4 EEG	∼0.6 M	0.613	0.565
	Sleep-EDF transfer	4 EEG	∼0.6 M	0.623	0.582
	Raw multimodal CNN	14 ch	0.95 M	0.655	0.612
Proposed feature-based multi-task model	MM-Net (neural only)	7 ch feat	0.77 M	0.721	0.635
	MM-Net (neural + cardio)	14 ch feat	0.86 M	0.722	**0.651**

Sleep-EDF, and a raw-signal convolutional network on all 14 channels, alongside the published single-channel baselines for the corpus [13]. We additionally evaluated a family of classical feature and ensemble pipelines (gradient boosting, a feature-sequence BiLSTM, a hemispheric graph-attention state-space model, and stacking and gradient-boosting ensembles); because these only confirm the staging ceiling discussed below, we report them in the text rather than tabulate them. All numbers are ten-fold patient-independent, at the hidden-Markov operating point where a model provides one.

6.4. Staging Benchmark and Comparison

Table 3 shows a clear pattern: every deep network, whether trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal, stays between 0.61 and 0.69 accuracy, 15 to 20 points below the same architectures' published accuracy on healthy sleepers. The compact, feature-based model reaches 0.722, above every deep baseline, which matches the concurrent finding that depth alone does not help on this cohort [16]: with under a hundred patients a convolutional network cannot rediscover what decades of sleep science already encode in the features. Staging accuracy is nonetheless capped. The strongest classical pipelines we evaluated reach 0.735 (a stacking ensemble) and 0.746 (a gradient-boosting ensemble), and

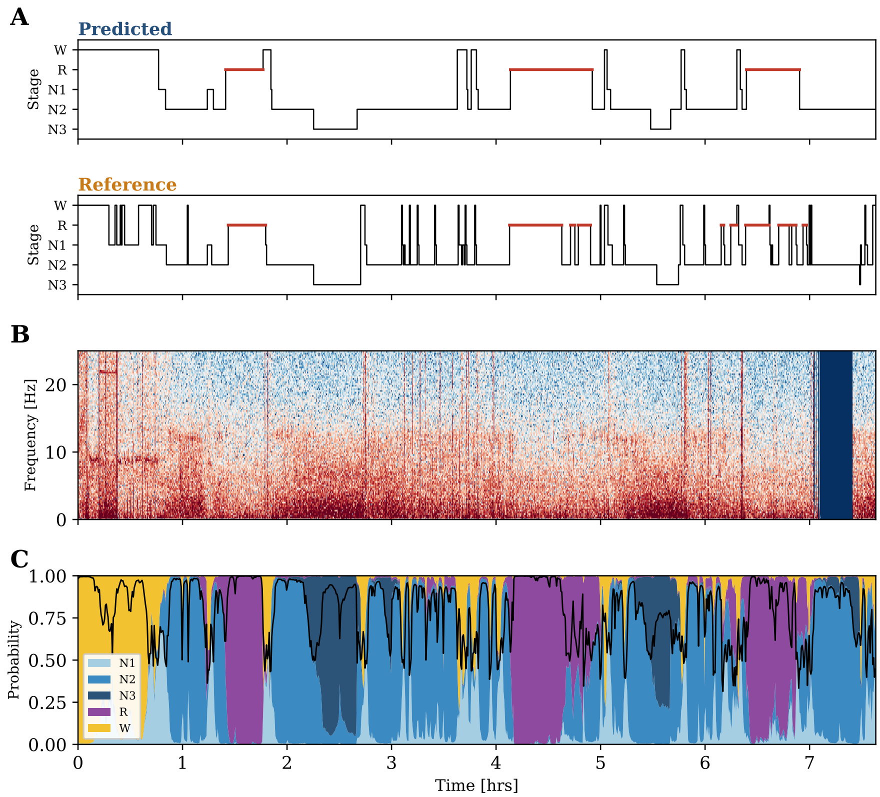

Figure 3: A representative held-out night (patient SN90, staging accuracy 0.80). **(A)** Predicted and reference hypnograms (REM highlighted). **(B)** EEG spectrogram (C4 derivation, 0–25 Hz). **(C)** The model's per-epoch stage-probability ribbon; the black line is the maximum posterior, i.e. its confidence. Deep sleep aligns with low-frequency spectral power, and confidence falls at stage transitions.

the published single-channel LSTM reaches 0.747; no method exceeds 0.75, so the task saturates on this cohort regardless of model family. We therefore do not position the model as a staging leader. What sets it apart is a second output: it is the only model that also detects respiratory events. A representative held-out night is shown in Figure 3: the predicted hypnogram follows the reference through the night's cycles, and the stage-probability ribbon exposes where the model is confident and where it hesitates.

6.5. Per-class Staging Performance

Table 4 breaks staging down by stage. The pattern is the one this cohort imposes on every method: N2, Wake, and REM are recovered well, N3 is

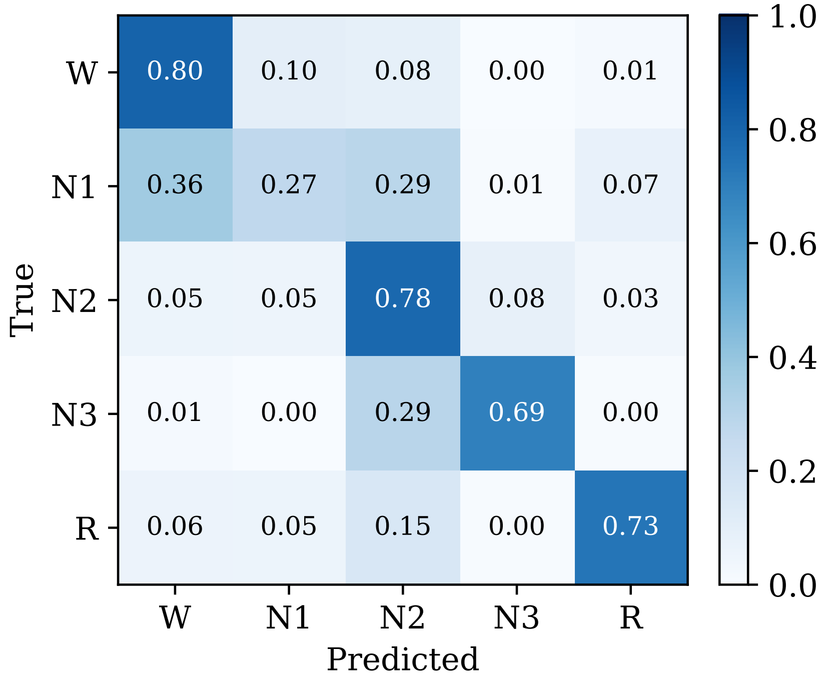

Figure 4: Row-normalised confusion matrix for the proposed model (pooled ten-fold test epochs, hidden-Markov operating point). Errors concentrate on N1, which is absorbed into Wake and N2.

Table 4: Per-class staging F1 (Wake, N1, N2, N3, REM), ten-fold patient-independent, at the hidden-Markov operating point, for the proposed model and the raw-signal deep network. Best per column in **bold**.

Model	W↑	N1↑	N2↑	N3↑	R↑
Raw multimodal CNN	0.74	**0.37**	0.73	0.56	0.67
MM-Net (neural only)	0.77	0.24	0.78	0.68	0.70
MM-Net (neural + cardio)	**0.77**	0.27	**0.78**	**0.68**	**0.72**

moderate, and N1 is the hard stage, because it is transient and rare. The raw convolutional network reaches the highest N1 F1, which fits its sensitivity to short events, while the proposed model leads on the stable stages. The row-normalised confusion matrix (Fig. 4) shows where the errors land: N1 is absorbed mainly into Wake and N2, and part of N3 is read as N2.

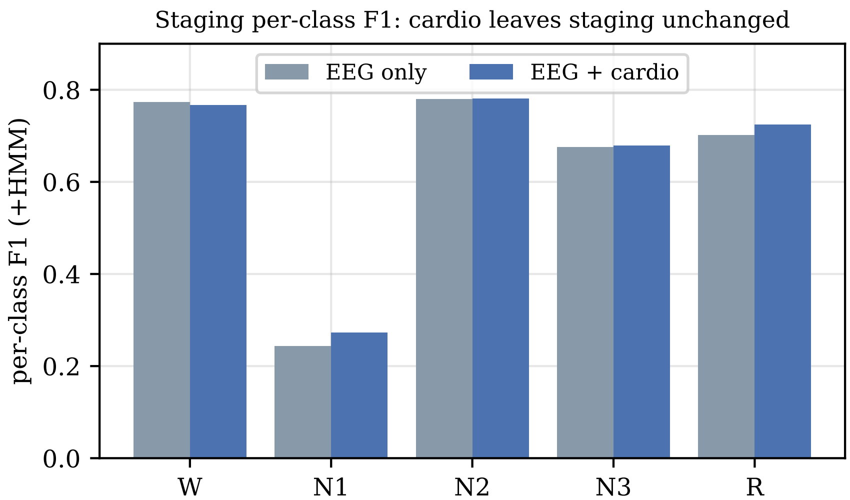

Figure 5: Per-class staging F1 for the neural-only and the neural-plus-cardiorespiratory model. Adding the cardiorespiratory channels leaves every stage at its full level, including Wake and REM.

6.6. Modality-Ablation Grid

This ablation is also our attribution method. Because the model is feature-based and never sees a raw spectrogram, a gradient heat-map or Grad-CAM would have no faithful spatial support; we therefore attribute each output by *causal* intervention — deleting a physiological channel and measuring the change — which is a falsifiable test of what each modality contributes rather than a post-hoc visualization. The proposed model stages sleep at 0.722 accuracy (κ 0.611) and detects respiratory events at 0.711 AUC (0.337 average precision). To test the paper's central claim, that each modality carries the task its physiology governs, we remove each modality in turn and keep the rest, on the same folds (Table 5). The pattern is clean. Removing SpO_2 drops respiratory AUC from 0.711 to 0.681 and leaves staging untouched, and removing the whole cardiorespiratory stream drops it furthest, to 0.673. Removing the ocular (EOG) channel does the reverse: staging falls from 0.722 to 0.712 while respiratory detection holds. A paired Wilcoxon test over the ten folds confirms the split: the cardiorespiratory stream changes respiratory AUC significantly (0.711 vs 0.673, $p = 0.004$) but not staging ($p = 0.91$). A cumulative build-up on the respiratory head shows SpO_2 alone already reaches the full AUC (0.711), so oxygen desaturation car-

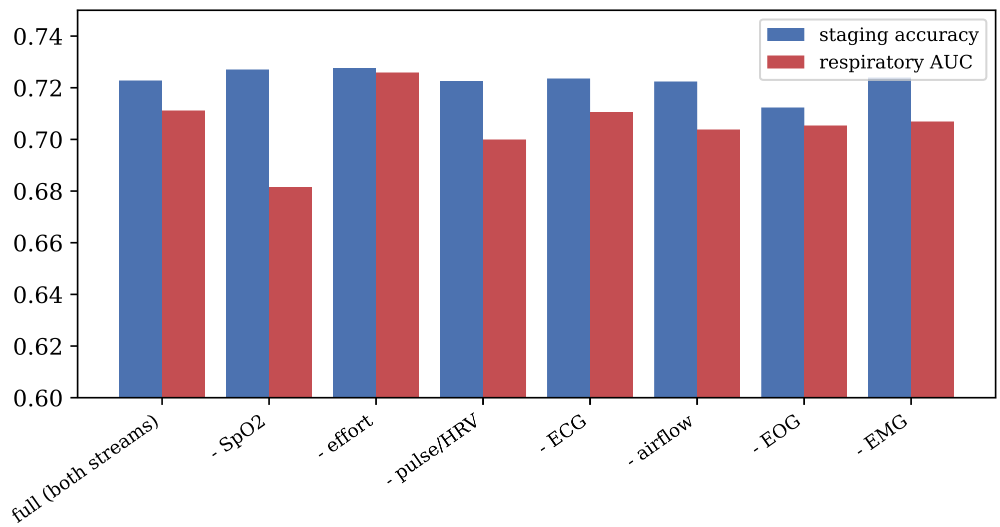

Figure 6: Modality-ablation grid. Staging accuracy (blue) and respiratory AUC (red) as each modality is removed in turn. Removing SpO_2 or the whole cardiorespiratory stream lowers respiratory AUC and leaves staging flat; removing the ocular channel lowers staging.

Table 5: Modality-ablation grid (leave-one-out), ten-fold patient-independent. Each row removes one modality and keeps the rest. Cardiorespiratory ablations (SpO_2, all cardiorespiratory) move respiratory AUC and leave staging flat; the ocular (EOG) ablation does the reverse. Largest effect per column in **bold**.

Modality removed	Staging acc ↑	Respiratory AUC ↑
none (full model)	0.723	0.711
SpO_2	0.727	**0.681**
respiratory effort	0.728	0.726
pulse / HRV	0.723	0.700
ECG	0.723	0.711
airflow	0.722	0.704
EOG (ocular)	**0.712**	0.705
EMG (muscle)	0.724	0.707
all cardiorespiratory	0.724	**0.673**

ries the read-out and the other channels are largely redundant for detection. Attention and plain concatenation stage and detect alike, so the gain comes from the added modality, and the fusion mechanism has little effect.

6.7. Physiological Validity of the Respiratory Read-out

Because the events were scored from airflow, a model that keyed on the airflow channel would be right for a trivial reason. Removing the airflow features leaves detection essentially unchanged, at 0.704 AUC (Table 5), within one fold standard deviation of the full 0.711. The model therefore does not lean on the labeling signal. It reads the fall in blood oxygen, the straining of the chest and abdomen, the beat-to-beat change in the heart, and the cortical arousal that ends the event. This is the useful regime, because it detects the bodily consequences of disordered breathing rather than repeating a measurement a technician has already made.

6.8. Respiratory Baselines and Per-event-type Detection

The respiratory task needs real competitors (Table 6), on the same folds and the same 14 cardiorespiratory features. A clinical desaturation rule reaches 0.596 AUC, logistic regression 0.582, and gradient boosting 0.670; a random classifier scores 0.160 average precision, the event prevalence. The proposed model reaches 0.711 AUC and 0.337 average precision, above gradient boosting but by a modest margin, so the contribution is the joint single pass and the physiological attribution above; we do not claim state-of-the-art detection accuracy. Detection also separates by event type: the model scores 0.692 AUC on hypopneas (the mild, dominant 81% of events), 0.763 on obstructive apneas, and 0.840 on central apneas. It is strongest on the most severe events, and the overall AUC is held down by the abundant hypopneas.

6.9. Clinical Validation against AHI and Severity

Aggregating the per-epoch respiratory probability into a per-patient burden and correlating it with the clinical apnea-hypopnea index (AHI) from the metadata gives a significant positive association (Spearman $\rho = 0.315$, $p = 0.002$, $n = 96$; Fig. 7), so the model's output tracks a clinician's summary measure rather than only an epoch-level score. Staging, in turn, degrades as sleep-disordered breathing worsens: accuracy falls from 0.770 in patients with a normal AHI to 0.708 in the severe group (Table 7). The pathology that motivates the second output also makes the first harder, which is the clinical reality of this cohort.

6.10. Model Complexity and Capability

Table 8 sets the proposed model beside representative alternatives on size and capability. A single compact network, under a million parameters

Table 6: Respiratory-event detection, ten-fold patient-independent. *(a)* Detectors trained on the 14 cardiorespiratory features on the same folds; chance-level average precision equals the event prevalence, 0.160. *(b)* The proposed model's AUC by event type – the rarer, more severe events are detected best. Best per column in **bold**.

(a) Detection method (14 cardiorespiratory features)

Method	AUC ↑	AP ↑
Desaturation rule	0.596	0.214
Logistic regression	0.582	0.193
Gradient boosting	0.670	0.290
MM-Net (proposed)	**0.711**	**0.337**

(b) MM-Net AUC by event type

Event type	Positive epochs	AUC ↑
Hypopnea	11,605	0.692
Obstructive	1,661	0.763
Central	831	**0.840**

Table 7: Sleep-staging accuracy by sleep-disordered-breathing severity (AASM AHI bands), ten-fold patient-independent; mean ± SD across patients ($N = 96$). Staging degrades monotonically as severity rises.

AHI band	AHI (events/h)	Patients	Staging acc ↑
Normal	< 5	14	0.770 ± 0.064
Mild	5–15	22	0.730 ± 0.077
Moderate	15–30	23	0.712 ± 0.103
Severe	≥ 30	37	0.708 ± 0.091

and an order of magnitude smaller than the raw-signal transformers, delivers both a competitive stage and a respiratory read-out from one pass over the recording. None of the staging-only models, at any size, can supply the second output.

7. Discussion

The result is a single model that reads the whole sleeping body and speaks to two clinical questions at once, in the population where both matter. The learned per-epoch embedding (Fig. 8) separates cleanly by sleep stage while

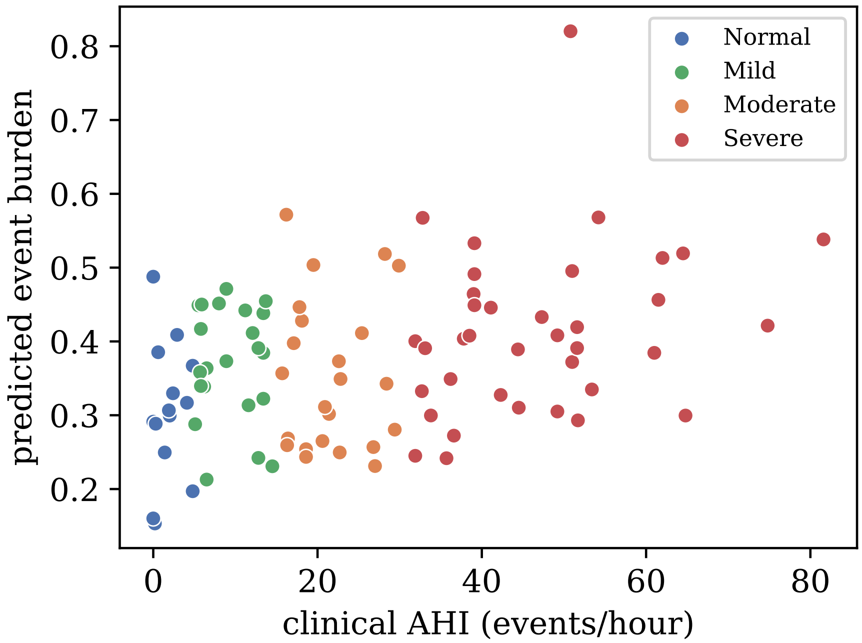

Figure 7: Predicted per-patient event burden (mean respiratory probability) against the clinical apnea-hypopnea index, coloured by AASM severity band. The positive association (Spearman $\rho = 0.315$, $n = 96$) shows the model's output tracks a clinician's summary measure.

Table 8: Model size and clinical capability. The proposed model is compact and is the only one that produces both a stage and a respiratory-event label.

Model	Input	Params↓	Outputs
CNN-ResNet18 [13]	1 EEG	∼11 M	stage
DeepSleepNet [2]	1 EEG	∼21 M	stage
Feature-sequence BiLSTM	7 ch feat	1.2 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	14 ch feat	**0.86 M**	**stage + apnea**

the respiratory structure stays diffuse, mirroring the strong staging and the modest respiratory AUC.

Staging accuracy on this cohort saturates below 0.75 for every model family we evaluated, and the proposed model reaches 0.722 while adding a respiratory capability the staging models lack. The respiratory read-out

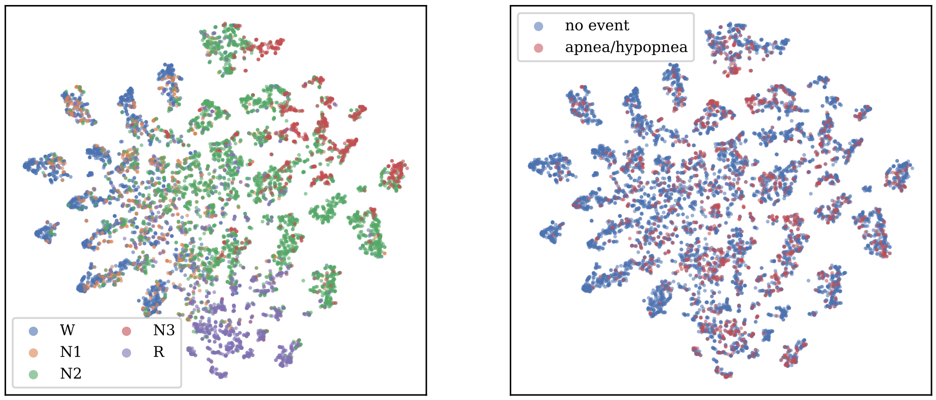

Figure 8: Two-dimensional t-SNE of the model's per-epoch embeddings (pooled test epochs). **Left:** coloured by sleep stage – Wake, N2, N3, and REM form distinct territories while N1 is diffuse. **Right:** coloured by respiratory-event label – events are spread across the manifold, consistent with the modest detection AUC.

does not depend on the labeling channel: it survives deletion of the airflow channel and rests on oxygen desaturation, effort, and arousal, the same signs a physician reads at the bedside.

The 0.72 staging figure should be judged on this cohort, where healthy-sleep accuracy does not transfer. Deep architectures developed on healthy sleepers collapse here: DeepSleepNet reaches 0.615, CNN-ResNet18 0.617, and a network pretrained on the healthy Sleep-EDF corpus and applied to this cohort 0.623, each roughly fifteen to twenty points below its healthy accuracy of about 0.85. The injured brain distorts the very micro-events, spindles, slow waves, and atonia, that those models learned to read. What holds up is physiologically grounded modeling on engineered features, at 0.72 to 0.75, which matches the published baseline on this corpus. The lesson is that healthy-sleep deep learning fails on the injured brain, and this cohort requires cohort-specific, physiologically grounded, multimodal modeling.

The study also draws a clean physiological line. The cortical channels carry sleep staging, and the cardiorespiratory channels carry breathing, and the model places each where it belongs rather than blurring them. That the neural and multimodal variants stage alike is the expected outcome: the second modality adds a new capability while leaving staging unchanged. We view this separation as a strength for clinical use, because a reader can trust that the respiratory output reflects breathing and the staging output reflects the brain.

7.1. Limitations

The evidence comes from a single center and the 96 patients with complete cardiorespiratory channels; external, multi-center validation is required before clinical use. Respiratory detection, at 0.711 AUC, is a useful signal but not yet a screening-grade one, and it is scored at the epoch level rather than per event. Staging sits at the cohort ceiling, so the model does not raise staging accuracy over strong classical pipelines. Finally, the model is feature-based: it inherits the assumptions of the engineered representations and is not an end-to-end raw-signal learner, a deliberate trade for data efficiency on a small cohort.

7.2. Future Work

Two directions follow naturally. Event-level metrics, such as sensitivity per event and false alarms per hour, would sit beside the epoch-level AUC and speak more directly to a sleep technician, and finer modeling of the breathing waveforms is likely to raise the respiratory number further. Linking the multimodal representation to stroke severity and recovery is the larger opportunity, and one the cardiorespiratory signal is well placed to serve.

8. Conclusion

We built a model that reads the full polysomnogram of stroke patients and, from one pass, stages their sleep and detects their respiratory events. Deep networks built for healthy sleep collapse to 0.61–0.69 accuracy on this cohort, so its 0.722 staging accuracy is at the cohort ceiling for every model family. The contribution is the second output, a respiratory read-out at 0.711 AUC that survives removal of the airflow channel, showing it reads the underlying physiology. Reading the whole recording gives the model a second clinical voice in the population that needs it most. Code and features are released.

Data and Code Availability

The iSLEEPS corpus is publicly available (Zenodo, DOI 10.5281/zenodo.14873844; and the india-data.org portal) as described by Maiti *et al.* [13]. Our code, engineered features, and reproduction notebook are released at `https://github.com/wshuv-o/isleeps-sleep-staging`.

Table 9: Recent single-channel sleep-staging methods: the accuracy each reports on healthy or general cohorts, set against its accuracy on the subacute-stroke cohort (iSLEEPS) for the two architectures we re-implemented. Healthy-cohort staging reaches 0.82–0.88; the same architectures fall to 0.62–0.69 on stroke, while the proposed multi-modal model reaches 0.722 on stroke and additionally detects respiratory events at 0.711 AUC (no staging method produces a respiratory output). A dash marks a value we did not evaluate.

Method	Year	Approach	Reported acc (dataset)	O
DeepSleepNet [2]	2017	Two representation-learning CNNs followed by a bidirectional LSTM sequence model, single-channel EEG.	0.82 (Sleep-EDF)	
AttnSleep [5]	2021	Multi-resolution CNN with multi-head self-attention over epochs, single-channel EEG.	0.844 (Sleep-EDF)	
SleepTransformer	2022	Transformer encoder with epoch-level interpretability and uncertainty estimates.	0.877 (SHHS)	
1D-ResNet-SE-LSTM	2023	Residual network with squeeze-and-excitation blocks and an LSTM on raw EEG.	0.864 (Sleep-EDF)	
Micro SleepNet	2023	Lightweight convolutional model for real-time mobile staging.	0.833 (SHHS)	
AT-BiLSTM	2024	Attention-augmented bidirectional LSTM, single-channel EEG.	0.838 (Sleep-EDF)	
MM-Net (proposed)	–	Two feature streams (neural and cardiorespiratory) fused into a BiLSTM that jointly stages sleep and detects respiratory events from the full polysomnogram.	–	

Ethics

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th IEC (BS&NS DIV.)/2022, dated 05.02.2022) [13]. This work is a secondary analysis of the de-identified public release and required no additional enrollment.

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